

MCY 612 Cloud Security Assignment 1

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Cloud Security

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In lab 1, you implemented some best practices. In this assignment, you will implement one of the following 3 security best practices in IAM. **You need to implement the assignment with AWS CLI.**

Configure a strong password policy for your users

https://docs.aws.amazon.com/IAM/latest/UserGuide/id_credentials_passwords_account-policy.html

https://docs.aws.amazon.com/IAM/latest/UserGuide/id_credentials_passwords_account-policy.html#PasswordPolicy_CLI

Report

Introduction

Using the AWS CLI to set up a strong password policy for AWS users is essential for preserving account security and guarding sensitive information. Organizations can lower the risk of unauthorized access resulting from weak or hacked passwords by implementing regulations such as minimum password length, complexity standards, expiration periods, and reuse limitations. Administrators can prevent brute force attacks and credential leaks by ensuring that users establish strong passwords that comply with security best practices by using the aws IAM update-account-password-policy command. Putting in place a clear password policy improves IAM security overall and assists in maintaining industry standards.

Materials

AWS Cloudshell terminal

AWS CLI

AWS Version

aws-cli/2.23.13 Python/3.12.6 Linux/6.1.127-135.201.amzn2023.x86_64 exec-env/CloudShell
exe/x86_64.amzn.2023

AWS Configure

AWS Access Key ID [*****FLZO2]

AWS Secret Access Key [*****23fZK]

Default region name [us-east-1]

Default output format [json]

Methods

The following code is used to create/ update a password policy, as well as implement strong security practice in order to protect user accounts.

```
~ $ aws iam update-account-password-policy --minimum-password-length 12 --require-symbols --require-numbers --require-uppercase-characters --max-password-age 90 --password-reuse-prevention 5
```

The following code is used to view the existing password policy.

```
~ $ aws iam get-account-password-policy
{
  "PasswordPolicy": {
    "MinimumPasswordLength": 12,
    "RequireSymbols": true,
    "RequireNumbers": true,
    "RequireUppercaseCharacters": true,
    "RequireLowercaseCharacters": false,
    "AllowUsersToChangePassword": false,
    "ExpirePasswords": true,
    "MaxPasswordAge": 90,
    "PasswordReusePrevention": 5
  }
}
```

Use the following code to delete the existing password policy.

```
~ $ aws iam delete-account-password-policy
```

Testing

Use the following code to check if there is a password policy in place.

```
~ $ aws iam get-account-password-policy
```

```
An error occurred (NoSuchEntity) when calling the GetAccountPasswordPolicy operation: The Password Policy with domain name 881490118823 cannot be found.
```

Conclusion

I used the aws IAM update-account-password-policy command to set up a strong password policy in AWS using the CLI. I specified parameters like the minimum password length, needed complexity such as uppercase, numerals, and symbols, password expiration, and password reuse prevention. This lowers the possibility of unwanted access because of weak or repeated passwords by guaranteeing that all IAM users adhere to stringent security requirements. Protecting key AWS resources, preventing brute force assaults, and ensuring compliance to industry standards and security best practices are all dependent on the implementation of strong password strategies.

References

Abelson, H., Anderson, R., Bellare, S. M., Benaloh, J., Blaze, M., Diffie, W., Gilmore, J., Green, M., Landau, S., Neumann, P. G., Rivest, R. L., Schiller, J. I., Schneier, B., Specter, M. A., & Weitzner, D. J. (2015, November 17). *Keys under doormats: Mandating insecurity by requiring government access to all data and communications*. OUP Academic. <https://academic.oup.com/cybersecurity/article/1/1/69/2367066>